3D ASSD per organ: boundary loss and 2.5D input

ResEnc DS U-Net · strong geometric augmentation · 50 epochs · 4-fold CV, 10 val patients per fold · 3 seeds per fold · lower is better

Esophagus

3D ASSD (mm)

1.73 ± 0.45 | 1.74 ± 0.39 | 1.73 ± 0.45 | 1.73 ± 0.44 | 1.69 ± 0.38

Heart

3D ASSD (mm)

2.46 ± 0.51 | 3.43 ± 1.40 | 2.22 ± 0.14 | 2.14 ± 0.04 | 2.48 ± 0.49

Trachea

3D ASSD (mm)

1.29 ± 0.18 | 1.49 ± 0.24 | 1.31 ± 0.11 | 1.33 ± 0.22 | 1.18 ± 0.14

Aorta

3D ASSD (mm)

1.54 ± 0.45 | 1.50 ± 0.29 | 1.49 ± 0.34 | 1.51 ± 0.42 | 1.48 ± 0.41

All organs (mean)

3D ASSD (mm)

1.75 ± 0.20 | 2.04 ± 0.30 | 1.69 ± 0.16 | 1.68 ± 0.15 | 1.71 ± 0.08

2D no boundary (control) | 2D ramped boundary | 2D fixed boundary 0.08 | 2.5D ramped boundary | 2.5D fixed boundary 0.08

One fold | Mean ± SD over the 4 folds | No boundary | Ramped boundary | Fixed boundary 0.08 | 2D input | 2.5D input (3 slices)

3D ASSD (mm) on the original CT grid at the best (val-selected) epoch; each point is a patient-averaged fold. Large marker = mean and SD over the 4 folds. Colour = loss, shape = input. One seed and 10 patients per fold: differences of a few thousandths of Dice are within noise. Y-axes differ by organ.